

Test 2: CS 3383 Theory of Automata

11:00 AM - 12:20 PM on 4/3/2025

Max. 100 points

Name:

R#:

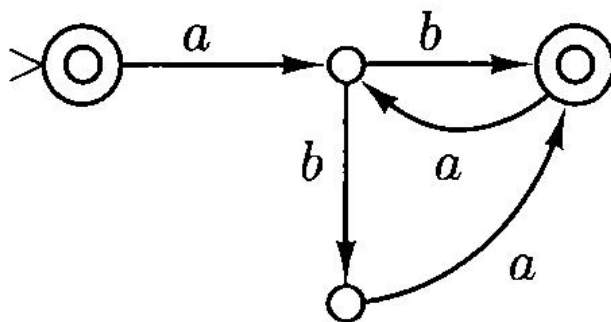
Signature:

This exam paper has two sides.

1. (30 points) Draw the state diagram of a nondeterministic finite automata that accepts

$$((a^*b^*a^*)^*b)^*$$

2. (30 points) Write down the regular expression for the language accepted by the nondeterministic finite automata

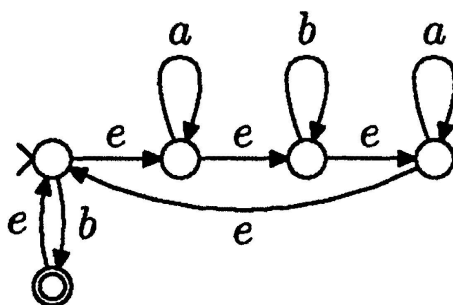


3. (40 points) Prove that the following language is not regular.

$$L = \{a^n b^n : n \geq 0\}$$

Test 2 solutions: CS 3383 Theory of Automata

1.



2. See problem 8 of assignment 2.

3. See page 109 of the Lecture Notes.